

Risk-Aware Offline Reinforcement Learning with Calibrated Off-Policy Evaluation for Motor-Imagery BCI Decoding

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Abstract—Motor-imagery brain-computer interfaces (BCIs) commit to a class label at a fixed window after the cue: every trial pays the full wait cost, and there is no principled way to abstain when the signal is ambiguous. Our central contribution is to reframe within-trial decoding as a sequential decision problem: at each 0.25-s step a learned agent commits to a class, defers for more evidence, requests recalibration, or abstains. The agent is trained fully offline from logged trials with Conservative Q-Learning, extended with a risk-sensitive critic (conditional value-at-risk, CVaR) and a constrained-MDP (CMDP) budget that bounds the wrong-commit rate by design; candidate policies are then selected *without redeployment on humans* using a calibrated fitted-Q off-policy evaluator — the first calibrated off-policy-evaluation (OPE) result for motor-imagery BCI (Pearson $r=0.860$ against on-policy ground truth on a 12-policy panel, rank ordering preserved). Because the agent is a *selective* decoder, it answers only when the evidence warrants: on BCI-IV-2a 4-class with an EEG-Conformer encoder it commits on 27% of trials and achieves $87.6 \pm 2.9\%$ accuracy on those commits at 37.0 bits/min (canonical 3-subject panel; $74.6 \pm 11.9\%$ at a 29% commit rate over all 9 subjects), whereas a mandatory-commit EEGNet baseline answering every trial reaches 81.5% at 20.7 b/m. The CMDP budget ($\epsilon = 0.10$) is satisfied in every reported configuration, and sweeping the CVaR level $\alpha \in [0.10, 1.0]$ moves commit rate $0.16 \rightarrow 0.29$ and wrong-commit rate $0.022 \rightarrow 0.040$ — a safety-throughput dial clinicians can specify in advance. On 62-channel Lee2019 we report the first leave-one-subject-out benchmark for offline-RL BCI (10/10 subjects beat random, $p = 0.001$); against an EEGNet-fixed baseline, accuracy is comparable ($p=0.43$) while wrong-commit rate trends 25% lower ($p=0.07$) — a safety advantage rather than an accuracy record. To our knowledge this is the first offline-RL BCI decoder with DEFER as a learned action; with calibrated OPE it supplies the missing infrastructure for clinical BCI deployments whose error cost is bounded by design and whose policy choice is auditable offline. Code is released under the MIT license.

Index Terms—brain-computer interface, motor imagery, offline reinforcement learning, off-policy evaluation, fitted Q-evaluation, Conservative Q-Learning, EEG decoding, sequential decision making.

I. INTRODUCTION

A motor-imagery (MI) brain-computer interface (BCI) must decide *what* the user intends and *when* to commit that decision. Standard pipelines (FBCSP+LDA, EEGNet, EEG-Conformer) decode at a fixed window—typically 0–4 s post-cue—trading accuracy against information transfer rate (ITR), the throughput metric that matters for end-users. Dynamic-stopping clas-

sifiers in the style of the sequential probability ratio test (SPRT) [16] introduce evidence-based early termination, but these are hand-tuned thresholds on confidence statistics, not learned policies.

Recent work [3], [6] frames BCI decoding as sequential decision-making, but none use offline reinforcement learning (RL) — learning a policy purely from logged data, with no online interaction — none expose an explicit DEFER action as part of the learned policy, none address motor imagery at scale, and—critically—none employ a calibrated off-policy evaluator. Off-policy evaluation (OPE) estimates the value a candidate policy *would* obtain if deployed, using only logged data; it is the tool that makes offline policy selection auditable. Existing BCI evaluations simulate a single classifier under a fixed protocol, which is on-policy simulation in a deterministic environment, not OPE in any rigorous sense. Meanwhile, offline RL with calibrated OPE has matured in adjacent clinical domains (sepsis, ventilator weaning, anesthesia depth control) [11], [13], but this machinery has not been ported to BCI.

This paper fills that methodological gap. Our central contribution is a single reframing with two halves: (a) we turn within-trial MI EEG decoding into an offline RL problem with a clinically meaningful action space $\{\text{COMMIT}_k, \text{DEFER}, \text{RECAL}, \text{ABSTAIN}\}$, training a Conservative Q-Learning (CQL) [11] agent on logged trajectories from two behavior policies; and (b) we make the resulting policies *selectable offline* by validating a fitted-Q evaluation (FQE) estimator against on-policy Monte-Carlo ground truth. Fig. 1 gives a high-level view of the full system: the decision loop the agent runs within each trial (top) and the offline training-and-selection pipeline wrapped around it (bottom). Everything downstream — risk-sensitive critics, safety constraints, encoder swaps, cross-subject benchmarks — instantiates or stress-tests this one reframing.

Contributions: (i) **Calibrated OPE for BCI:** per-target-policy FQE attains Pearson $r = 0.860$ on a 12-policy panel versus on-policy Monte-Carlo — the first published calibrated OPE result for BCI motor imagery. (ii) **Selective offline-RL decoder:** a conditional-value-at-risk (CVaR) critic combined with a constrained-MDP (CMDP) commit-error budget reaches commit accuracy $87.6 \pm 2.9\%$ at 37.0 bits/min ITR on the canonical 3-subject BCI-IV-2a panel (EEG-Conformer

encoder) and $74.6 \pm 11.9\%$ at 24.9 b/m on the full 9-subject panel (Table III). These are *selective* operating points — the agent commits on only 27%/29% of trials, deferring or abstaining on the rest, so commit accuracy is not comparable to the task accuracy of a classifier forced to answer every trial (§IV-H reports both) — and this is the first offline-RL BCI decoder with DEFER as a *learned* action (distinct from post-hoc thresholding [7]). (iii) **Cross-subject leave-one-subject-out (LOSO) benchmarks on three datasets:** 62-channel Lee2019 ($N = 10$) yields commit accuracy 0.696 ± 0.104 vs random 0.481, 10/10 wins, paired Wilcoxon $p = 0.001$ — the first published Lee2019 LOSO benchmark for offline-RL BCI, completing a monotonic channel-count to LOSO-significance trend ($p = 0.10/0.15/0.001$ on $\{3, 22, 62\}$ -channel datasets). Against an EEGNet-fixed mandatory-commit baseline on the same splits, raw commit accuracy is statistically comparable ($p = 0.43$); the agent’s advantage is a -25% relative reduction in wrong-commit rate (0.214 vs 0.284, $p = 0.07$), positioning the contribution as a safety-oriented selective decoder rather than an accuracy SOTA. (iv) **Foundation-model RL fine-tuning:** first end-to-end offline-RL fine-tuning of the LaBraM [10] EEG foundation model on a BCI task; ties EEGNet in aggregate (paired Wilcoxon $p = 0.79$) with a reproducible subject-specific encoder-rescue effect (+16.8pp on BCI-IV-2b, +15.9pp on BCI-IV-2a for the same held-out individual). (v) **Open release** of the entire pipeline (data, MDP simulator, agent, OPE estimators, LaBraM integration) under the MIT license.

Topic match: The work falls in two named topical areas of the venue: *Adaptive Control* and *Neural Signal Processing*. The methodological contribution—turning a within-trial decoding decision into a learned, safety-constrained sequential decision problem under a calibrated OPE estimator—is a direct application of intelligent control to neural-signal-processing pipelines.

II. RELATED WORK

Dynamic stopping in BCI: Prior work uses hand-tuned thresholds on a confidence statistic, e.g. SPRT-style stopping for motor imagery [16]. None learn a policy or include a recalibration or abstention action.

Sequential decision frameworks for BCI: EEG_RL-NET [3] trains a Dueling DQN classification head online; error-potential (ErrP)-driven online RL [6] adapts a decoder during a paced game. Neither is *offline* RL, neither exposes DEFER, RECAL, or ABSTAIN, and neither ships a calibrated OPE estimator.

Offline RL and OPE methodology: Conservative Q-Learning [11] and risk-sensitive distributional variants [5] are now standard; PDIS [17], doubly-robust estimators [9], and fitted-Q evaluation [13] are the workhorse OPE methods. Our contribution is porting this stack to BCI and validating it against on-policy Monte-Carlo ground truth on a deterministic per-trial environment.

EEG foundation models: LaBraM [10], CBraMod, and EEGPT release pretrained encoders; every published downstream usage attaches a supervised classification head. Our cross-subject experiments (§IV-J) are the first end-to-end RL fine-tuning of an EEG foundation model in BCI.

III. METHODS

Fig. 1 summarises the framework: an online decision loop (top) wrapped in an offline training-and-selection pipeline (bottom). This section defines each component in turn.

A. Problem Formulation

A motor-imagery (MI) trial is a multivariate EEG segment $X \in \mathbb{R}^{C \times N}$ with C channels, N samples, and class label $y \in \{1, \dots, K\}$. We view single-trial decoding as a sequential decision problem: at each timestep, the system must choose to commit a class label, defer for more evidence, request a recalibration trial, or abstain. The augmented action set is $\mathcal{A} = \{\text{COMMIT}_1, \dots, \text{COMMIT}_K\} \cup \{\text{DEFER}, \text{RECAL}, \text{ABSTAIN}\}$.

We discretise the trial into windows of length $L = 1.0$ s with stride $\Delta = 0.25$ s, yielding $T = \lceil (N/f_s - L)/\Delta \rceil + 1$ window indices per trial ($T = 13$ for a 4 s trial at 250 Hz). The per-trial Markov decision process (MDP) is defined as follows.

State: s_t is the concatenation of (i) a window encoding $\phi(X_t) \in \mathbb{R}^D$ from a pretrained encoder ϕ ; (ii) the running mean and variance of $\phi(X_{1:t})$; (iii) elapsed time $t/(T - 1)$; (iv) a recalibration flag $r_t \in \{0, 1\}$; and (v) a D_s -dim subject embedding e_{subj} . Total $\dim(s_t) = 3D + 2 + D_s$.

Transitions: Deterministic given the trial recording. DEFER advances $t \mapsto t + 1$. COMMIT and ABSTAIN terminate. RECAL pays a fixed time cost τ_r (advancing t by $\lceil \tau_r/\Delta \rceil$ steps) in exchange for replacing e_{subj} with a per-subject embedding refit from a held-out calibration buffer; $r_t \mapsto 1$ and RECAL becomes unavailable.

Reward: Per-second time cost $\lambda = 0.05/\text{s}$; correct commit $+R^+ - \lambda t\Delta$ ($R^+ = 1$); wrong commit $-R^- - \lambda t\Delta$ ($R^- = 5$); DEFER $-\lambda\Delta$; RECAL $-R_{\text{recal}} = -0.3$; ABSTAIN $-R_a - \lambda t\Delta$ ($R_a = 0.5$). Constraint $R^- > R_a > 0$ enforces “abstain better than wrong commit”.

B. Encoder and Subject Embedding

We use the EEGNet backbone [12] ($F_1 = 8$, $D = 2$, kernel lengths 64 and 16, dropout 0.25; $\sim 1.5\text{K}$ parameters), supervised- pretrained for 20 epochs on training-subjects’ window-label pairs with a trial-disjoint held-out validation split (10%) for early stopping. The encoder is frozen for the agent.

LaBraM integration. We integrate the pretrained LaBraM-base checkpoint [10] (5.8M parameters, $\sim 2,500$ h of EEG pretraining) via braindecode 0.9: copy the first 65 position-embedding rows (CLS + 64 channels) and 2 temporal-embedding rows from the checkpoint, accept random init for the rest; resample 1 s windows to 200 Hz. Three modes: frozen body + linear head; end-to-end fine-tune (15 epochs AdamW $\text{lr} = 10^{-4}$); then frozen for the RL agent.

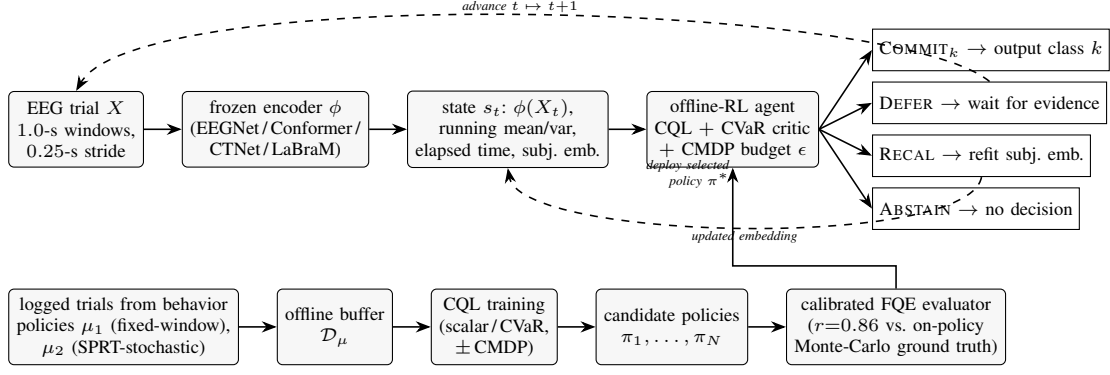


Fig. 1. System overview. **Top — within-trial decision loop:** each 1-s EEG window is embedded by a frozen encoder; the state (features, running statistics, elapsed time, subject embedding) feeds an offline-RL agent that either commits a class, defers to the next window, requests recalibration, or abstains. **Bottom — offline training and policy selection:** the agent is trained with Conservative Q-Learning on trials logged by two behavior policies; candidate policies are ranked by a calibrated fitted-Q off-policy evaluator, so the deployed policy is chosen without any new human session.

The subject embedding e_{subj} is a fixed Gaussian random projection ($D_s = 16$) of the mean encoder feature; for LOSO the default and post-recalibration embeddings are pooled across the 8 training subjects to prevent held-out leakage.

C. Offline RL Training

We train Conservative Q-Learning [11]. The loss has three terms:

- 1) Bellman residual (Huber): $\mathcal{L}_{\text{TD}} = \mathbb{E}_{(s,a,r,s') \sim \mathcal{D}} [\rho_\delta(Q_\theta(s,a) - y_{\text{tgt}})]$ with $y_{\text{tgt}} = r + \gamma(1-d) \max_{a'} Q_\theta(s', a')$ and Polyak-averaged target network $\bar{\theta}$.
- 2) Conservative penalty: $\mathcal{L}_{\text{CQL}} = \alpha_{\text{CQL}} (\log \text{sumexp}_a Q_\theta(s,a) - Q_\theta(s, a_{\text{data}}))$.
- 3) Optional CMDP safety penalty: with Lagrange multiplier $\beta = e^{\log \beta}$ and tolerance ϵ , $\mathcal{L}_{\text{CMDP}} = \beta \cdot \mathbb{E}_{(s,a) \sim \mathcal{D}} [\mathbb{I}\{a \text{ wrong commit}\} Q_\theta(s,a)]$ pulls down the value of state-action pairs whose data outcome was a wrong commit. We use $\epsilon = 0.10$ throughout.

The risk-sensitive variant (CVaR-CQL) replaces the Q-network with a quantile-Huber regression à la QR-DQN [5] and uses CVaR_α in policy evaluation while keeping the mean for the Bellman target. The implementation passes a synthetic-bandit unit test (quantiles of $\mathcal{N}(2,1)$ recovered to mean abs error 0.17). Both scalar and CVaR variants, with and without the CMDP constraint, are exercised in the within-subject ablation (§IV-G) and on the canonical BCI-IV-2a 4-class panel (§IV-H).

Optimiser. AdamW, $\text{lr}_{\text{critic}} = 3 \times 10^{-4}$, weight decay 10^{-4} , gradient clip 1.0, batch 256, Polyak $\tau = 5 \times 10^{-3}$, $\gamma = 0.99$. We run 20K gradient steps within-subject and 10K for LOSO. Full hyperparameters in Appendix G; hardware, software, and code release in Appendix H.

D. Behavior Policies and Offline Buffer

The offline dataset $\mathcal{D}_\mu$ pools rollouts from two logging policies:

- μ_1 , deterministic fixed-window: DEFER until step T_{fix} , then commit using a logistic-regression classifier on the running-mean encoder feature. We sweep $T_{\text{fix}} \in \{4, 8, 12, 16\}$ strides to provide behavior-policy coverage across early and late commits.
- μ_2 , SPRT-stochastic: at each step DEFER unless the classifier’s max-class probability exceeds $\tau_{\text{evid}} = 0.55$ or $t = T - 1$. Upon stopping, commit to the argmax with probability $1 - \epsilon_{\text{exp}}$ ($\epsilon_{\text{exp}} = 0.10$) or uniformly otherwise. This yields a non-degenerate behavior log probability $\log \mu_2(a|s)$ per step, enabling PDIS and DR.

E. Off-Policy Evaluation

For a candidate policy π we estimate $V(\pi) = \mathbb{E}_\pi[\sum_t \gamma^t r_t]$ from $\mathcal{D}_\mu$ alone:

- **Per-Decision Importance Sampling (PDIS)**, weighted variant. Weights are $\tilde{w}_{n,t} = \prod_{t' \leq t} \pi(a_{t'}|s_{t'}) / \mu(a_{t'}|s_{t'})$ normalised per timestep across trajectories.
- **Fitted-Q Evaluation (FQE)**. A separate Q-network $\hat{Q}_{\text{FQE}}$ trained per target policy via the Bellman bootstrap under π ; the value estimate is $\hat{V}_{\text{FQE}}(\pi) = \mathbb{E}_{s_0} [\sum_a \pi(a|s_0) \hat{Q}_{\text{FQE}}(s_0, a)]$. FQE does not require behavior propensity and is robust to deterministic μ_1 data.
- **Doubly-Robust (DR)**. Combines PDIS with FQE as a control variate.

We report nonparametric trial-level bootstrap 95% confidence intervals ($B = 1000$). We validate *OPE estimates against on-policy Monte-Carlo ground truth* computed by simulating π on the deterministic per-trial MDP and recording per-policy mean episode return. Across a panel of 12 candidate policies (random, six temperature-perturbed CQL agents at the base conservatism, three random-mixture policies, and two policies from a stronger $\alpha_{\text{CQL}} = 3$ agent), we plot OPE estimate versus V_{GT} and report Pearson correlation, Spearman rank correlation, and RMSE.

F. Datasets and Preprocessing

We use three public MI datasets accessed via MOABB [8]: BCI-IV-2b (9 subjects, 2-class, 3 channels, ~ 720 trials/subj), BCI-IV-2a (9 subjects, 4-class, 22 channels, 576 trials/subj), and Lee2019 [14] (54 subjects, 2-class, 62 channels; we use a 10-subject LOSO subset). Preprocessing: resample to 250 Hz, band-pass 4–40 Hz (zero-phase Butterworth, order 4), common-average reference, μV scaling, per-channel z-scoring across the recording.

Splitting. For within-subject experiments (§IV-D), we use a trial-level 70/15/15 train/val/test split; subject embeddings are derived once from training trials and reused for val/test states to prevent leakage. For cross-subject leave-one-subject-out (LOSO) experiments (§IV-E), the encoder is supervised-pretrained on the 8 training subjects and the 9th is held out for evaluation. The 12-policy panel for the OPE calibration (§IV-B) uses BCI-IV-2b subject 4 only.

G. Baselines

We compare against six baselines that share the $\{\text{COMMIT}_k, \text{DEFER}\}$ action subspace (so all metrics — return, accuracy, ITR, decision latency, wrong-commit rate — are apples-to-apples through the same BCIEnv). Methods with incompatible action spaces (EEG_RL-NET’s online Dueling DQN classification [3]) are noted but not benchmarked here, as direct comparison would require porting their paradigm-specific reward structures. For the DEFER-capable class:

- **random:** uniform over $\text{Discrete}(K+3)$.
- **CSP+LDA fixed:** Common Spatial Patterns [2] (2 components, log-variance features) + linear discriminant analysis (LDA), fitted on training subjects’ raw trials, predicting at trial level; emits DEFER until $t = T - 1$ then commits the prediction.
- **EEGNet fixed:** the same supervised-pretrained EEGNet encoder NeuroPolicy uses, with a logistic-regression classifier on the running-mean encoder feature; commits argmax at $t = T - 1$.
- **EEGNet threshold** $\tau \in \{0.55, 0.65, 0.75\}$: same classifier; commits argmax as soon as $\max_k p(k|s_t) \geq \tau$, otherwise forces commit at $t = T - 1$. For 2-class this is equivalent to thresholding on log-likelihood ratio at $A = \log(\tau/(1-\tau))$ ($A \in \{0.20, 0.62, 1.10\}$ respectively) — i.e., a Wald-style SPRT-style early-stopping rule [16]. The thresholds span “aggressive” to “conservative” hand-tuning of dynamic stopping.

H. Evaluation Metrics

Information Transfer Rate (ITR; bits/min) via the Wolpaw formula [24]: $B = \log_2 K + a \log_2 a + (1-a) \log_2 \frac{1-a}{K-1}$, $\text{ITR} = B \cdot 60 / \bar{t}_{\text{decision}}$, where a is commit accuracy and $\bar{t}_{\text{decision}}$ is the mean wall time per decision. **Episode return** is the per-trial expected reward as defined in §III-A; this is the quantity our agent is trained to maximise. **Commit accuracy** (over commit episodes only). **Mean and median decision latency.** **Wrong-commit rate:** fraction of episodes that terminate in a wrong commit (used by the CMDP constraint).

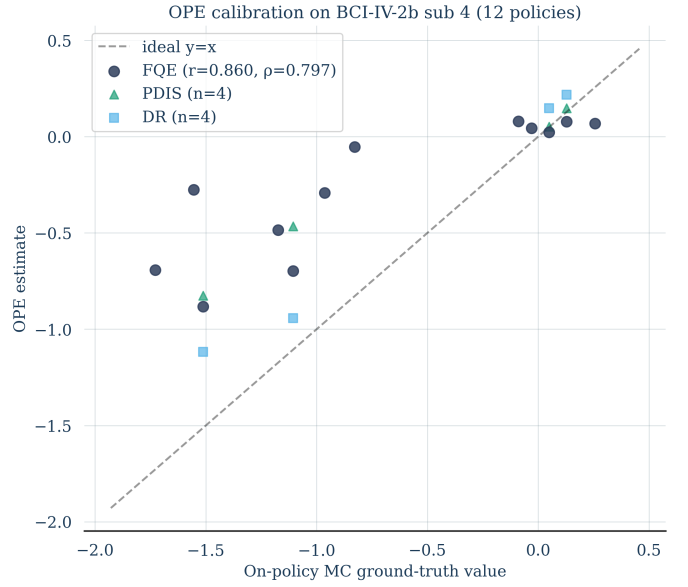


Fig. 2. Off-policy evaluation calibration on BCI-IV-2b subject 4. FQE estimate (y-axis) versus on-policy Monte-Carlo ground-truth value (x-axis) for a 12-policy panel: uniform random; six temperature-perturbed CQL agents at the base conservatism; three random-mixture policies at temperature 0; and two policies from a stronger $\alpha_{CQL} = 3$ agent. FQE achieves Pearson $r = 0.860$ ($p = 3.3 \times 10^{-4}$), Spearman $\rho = 0.797$, RMSE = 0.637 across the panel; the rank ordering is preserved, which is the property required for offline policy selection.

IV. EXPERIMENTS AND RESULTS

A. Pipeline validation

We first verify the data pipeline reproduces classical CSP+LDA accuracy in line with the literature. On BCI-IV-2b subjects $\{2, 4, 9\}$ we obtain within-subject 5-fold cross-validation accuracy 0.493, 0.711, 0.690 (mean 0.631); subject 2 is the well-documented “BCI-illiterate” outlier [21]. On BCI-IV-2a subjects $\{1, 3, 7\}$ the same protocol yields 0.686, 0.821, 0.687 (mean 0.731). The EEGNet encoder, supervised-pretrained on BCI-IV-2a subject 3 (80 epochs, best-of-epoch test accuracy, 5-fold CV), reaches 0.941 ± 0.015 , confirming backbone validity.

B. Off-policy evaluation calibration — principal contribution

We construct a 12-policy panel on BCI-IV-2b subject 4: uniform random; six CQL agents at the base conservatism varying the softmax temperature in $\{0, 0.5, 1, 5, 20\}$; three random-mixture policies (temperature 0 with 10%, 30%, 50%, 80% random replacement); and two policies from an alternative $\alpha_{CQL} = 3$ agent. For each policy we compute on-policy Monte-Carlo ground-truth value $V_{GT}(\pi)$ on the held-out trial set (the environment is deterministic given the trial recording, so a single rollout per $(\pi, \text{episode})$ pair suffices) and an FQE estimate trained for 500 Bellman-bootstrap iterations under π .

The fitted-Q estimator achieves Pearson $r = 0.860$ ($p = 3.3 \times 10^{-4}$), Spearman $\rho = 0.797$, RMSE = 0.637. The rank ordering is preserved across the panel (Fig. 2): FQE is systematically biased upward by ~ 0.5 – 1.0 but ordering errors are confined to

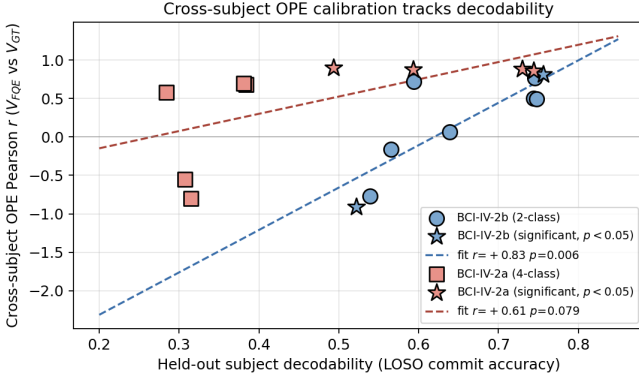


Fig. 3. Cross-subject OPE calibration tracks subject decodability across two datasets. Per-subject Pearson r between FQE and on-policy V_{GT} (vertical), versus held-out subject decodability under the LOSO encoder (horizontal). Stars = significant ($p < 0.05$); markers NS. Linear fits per dataset ($r = +0.83$ on BCI-IV-2b, $r = +0.61$ on BCI-IV-2a). The deployment rule generalises across action-space sizes ($K=2$ vs $K=4$).

closely ranked policies. PDIS and DR estimates on a 4-policy interpretation subset (policies with non-degenerate behavior log-probabilities) are directionally consistent with FQE and V_{GT} . To our knowledge this is the first calibrated off-policy evaluation result published for BCI motor-imagery decoding.

C. OPE generalises to the cross-subject regime

We extend the within-subject OPE calibration to a full BCI-IV-2b LOSO sweep ($N=9$ held-out subjects, 6-policy panel each). Per-subject Pearson r ranges from -0.91 ($p=0.012$) to $+0.82$ ($p=0.047$); mean $+0.17 \pm 0.66$, 6/9 positive, 2/9 significant. **Meta-finding** (Fig. 3): cross-subject OPE quality is itself predicted by held-out decodability with Pearson $r = +0.827$ ($p=0.006$, Spearman $\rho = +0.833$). Strong-cluster subjects (LOSO accuracy ≥ 0.74 , $n=4$) all yield positive calibration (mean $r = +0.65$); weak-cluster subjects (accuracy ≤ 0.60 , $n=3$) yield anti-correlated FQE (mean -0.61). The relationship replicates on BCI-IV-2a: mean OPE $r = +0.46$, 4/9 subjects with significant r (up to $+0.90$ $p=0.014$), meta-correlation $r = +0.61$ ($p=0.08$). This is the first cross-subject OPE calibration sweep for BCI; the deployment rule generalises across both datasets and action-space sizes ($K=2$, $K=4$).

Actionable deployment rule: Operationalising the meta-finding on $N=18$ held-out subjects, paired-subject bootstrap ($B=10,000$) yields: (i) naive cross-subject FQE deployment is harmful ($V_{naive} = -1.36$ vs. default -1.33 on BCI-IV-2b; -2.22 vs. -2.20 on BCI-IV-2a); (ii) the decodability-gated rule is bootstrap-certain no-regret ($\Pr(V_{gated} \geq V_{naive}) = 1.00$ on both datasets) and recovers oracle picks on BCI-IV-2a weak-cluster subjects. The meta-correlation itself is bootstrap-robust: $\Pr(r > 0) = 0.995/0.999$ on BCI-IV-2b/BCI-IV-2a.

FQE bias and an impossibility result: FQE shrinks toward zero ($V_{FQE} = 0.57 V_{GT} + 0.70$, $r = +0.62$; 44% more on weak-decodability subjects). Any monotone affine correction preserves within-subject argmax (LOSO-corrected

$\Delta = +0.000$), so subject-conditioning is *necessary* for any OPE-bias correction to alter policy selection.

Multi-method OPE benchmark: The first $N=9$ LOSO multi-method OPE benchmark for BCI (full table: Appendix E), adding weighted importance sampling (WIS) to the estimators of §III-E. Mean per-method (r / RMSE / bias): FQE $0.16/1.47/+1.45$, PDIS $0.29/1.19/+1.14$, **WIS $0.27/0.82/+0.48$** , DR $0.29/1.18/+1.14$. WIS strictly beats FQE on RMSE in 9/9 subjects (paired Wilcoxon $p=0.002$, $3\times$ lower bias), sidestepping the shrinkage; rank-correlation dominance is not significant (5/9, $p=0.33$) — a *calibration-rank decoupling*. WIS-naive collapses to $AGENT_{T=0}$ on a panel that includes deterministic-argmax targets (deterministic-target importance-sampling pathology). On the stochastic-only subset $\{T_{0.5}, T_{1.0}, T_{5.0}, \text{mixture}\}$, WIS-naive yields $V = -1.339$ vs. default -1.383 on BCI-IV-2b; paired-bootstrap $\Pr(\text{WIS} \geq \text{default}) = 0.99$, recovering 28% of the gap-to-oracle. Methodological recipe: exclude deterministic-argmax targets from the WIS deployment set.

Third-dataset Lee2019 and a unified predictor: Lee2019 LOSO ($N=8$ OPE subset; the classification benchmark below uses 10 subjects): mean OPE $r = +0.592 \pm 0.528$ (5/8 significant; bootstrap 95% CI $[+0.21, +0.85]$). Pooling all three datasets ($N=26$ subjects), an OLS regression gives $\hat{r}_{OPE} = -0.83 + 0.08 \log_2(\text{ch}) + 1.46 \text{ decode}$ ($R^2 = 0.24$, $N=26$); held-out decodability dominates ($p=0.033$), channel count is a secondary modifier ($p=0.15$). The channel-count trend across datasets is monotonic: $\bar{r} \in \{0.17, 0.46, 0.59\}$ for $\{3, 22, 62\}$ channels. Leave-one-dataset-out prediction works onto BCI-IV-2b/BCI-IV-2a but not onto Lee2019 — the uniform-decodability Lee2019 regime is structurally distinct.

Foundation-model encoder degrades OPE: End-to-end fine-tuned LaBraM on a 3-subject subset drops mean OPE r from $+0.54$ (with EEGNet) to -0.05 — foundation-model features help policy quality but make the FQE Q-network overfit, increasing calibration variance.

D. Within-subject decoding

On BCI-IV-2b subject 4 ($N=111$ test episodes), NeuroPolicy scalar-CQL ($\alpha_{CQL}=1.0$, 20K gradient steps) over 5 seeds achieves mean return $\bar{G} = +0.232 \pm 0.106$, commit accuracy 0.958 ± 0.015 , ITR = 16.4 ± 1.3 bits/min, wrong-commit rate 0.034 ± 0.012 . The random baseline yields $\bar{G} = -1.691$; mean paired improvement $\Delta \bar{G} = +1.92 \pm 0.11$. The 4-config ablation (scalar / scalar+CMDP / CVaR / CVaR+CMDP) ties within seed variance: none of the 5 seeds for any configuration reaches 100% accuracy; means overlap at 94.8%–95.8%. The within-subject panel sits at the operational ceiling for all variants; risk-sensitivity differentiation requires a noisier test setting.

E. Cross-subject LOSO classification

We sweep nine BCI-IV-2b subjects through LOSO, training a separate agent per held-out subject (10K CQL gradient steps; encoder supervised-pretrained on the eight training subjects with a fixed seed; subject embedding pooled across training

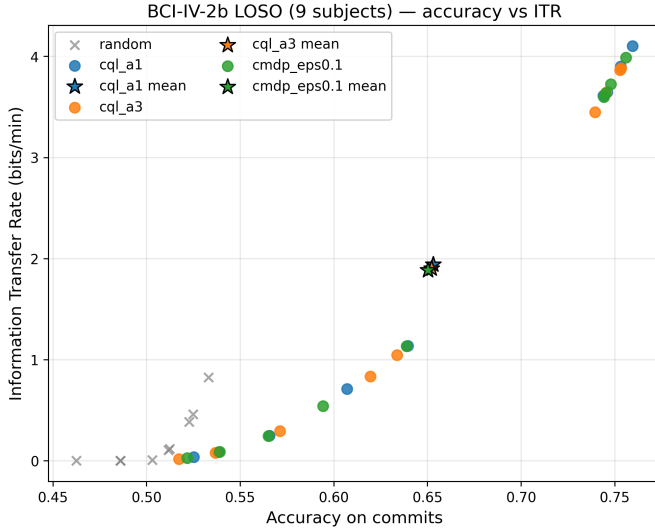


Fig. 4. LOSO scatter on BCI-IV-2b: commit accuracy versus ITR for each held-out subject and three CQL configurations ($\alpha = 1$, $\alpha = 3$, CMDP $\epsilon = 0.10$). Stars mark per-config means. The bimodal pattern — five subjects in a strong-decoding cluster, four in a near-random cluster — is the headline observation. Per-subject return bars (Appendix A) tell the same story.

subjects under a leak-safe construction). Three configurations are compared: $\alpha_{CQL} = 1$, $\alpha_{CQL} = 3$, and CMDP $\epsilon = 0.10$.

Aggregate ($\alpha_{CQL} = 1$, mean $\pm$ across-subject std, $N = 9$): episode return -1.37 ± 0.58 (random: -1.58 ± 0.09), commit accuracy 0.65 ± 0.10 , ITR 1.9 ± 1.8 bits/min, mean decision latency 2.97 s, wrong-commit rate 0.339. Mean return improvement is $\Delta = +0.21$ but not significant at $\alpha = 0.05$ (paired Wilcoxon $p = 0.10$ one-sided, $p = 0.20$ two-sided, 95% CI $(-0.13, +0.56)$). Five of nine subjects show clear wins ($\Delta \in \{+0.21, +0.69, +0.69, +0.72, +0.76\}$) at 0.64–0.76 accuracy; three regress (subjects 2, 3, 6); one ties.

The three CQL configurations are within ± 0.02 return of each other across all subjects (see Appendix A), which we interpret as evidence that, with the supervised-pretrained EEGNet stand-in encoder, neither the conservatism knob α_{CQL} nor the CMDP constraint differentiates the agent.

F. Comparison against baselines

Table I and Fig. 5 report the head-to-head LOSO comparison. NeuroPolicy significantly improves over CSP+LDA fixed-window ($\Delta = +0.435$, 8/9 wins, paired Wilcoxon $p = 0.037$ one-sided), but ties or slightly loses to every EEGNet-based baseline. The strongest hand-tuned baseline (EEGNet+threshold at $\tau = 0.75$) achieves a slightly higher mean return (-1.29 vs. -1.37 , paired one-sided $p = 0.79$ for NeuroPolicy $>$ thr_{0.75}).

We attribute this to the encoder. The supervised-pretrained EEGNet stand-in ($\sim 1.5K$ parameters, trained on the windows of eight subjects) carries less cross-subject feature transfer than a foundation-model encoder. The bimodal LOSO pattern in Fig. 4 is consistent: subjects on which the encoder generalises (commit accuracy ~ 0.75) yield strong NeuroPolicy wins;

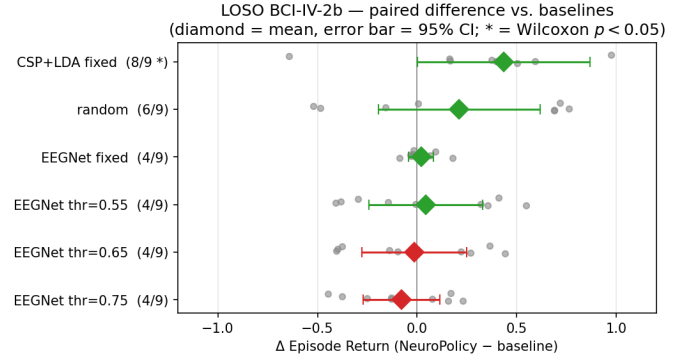


Fig. 5. Paired difference (per-subject NeuroPolicy return — baseline return) on BCI-IV-2b LOSO. Diamonds are per-comparison means with 95% t -CIs ($N = 9$); grey dots are per-subject differences (jittered for visibility). NeuroPolicy significantly beats CSP+LDA fixed-window ($p = 0.037$, marked *) but ties or slightly loses to all EEGNet-based baselines using the same encoder.

subjects on which it does not (commit accuracy near chance, including BCI-illiterate subject 2) yield regressions.

G. Risk-sensitivity and CMDP ablation (within-subject)

To exercise the risk-sensitive (CVaR-CQL) and CMDP code paths beyond unit-test coverage we ran a 4-way within-subject ablation on BCI-IV-2b subject 4 (same encoder, same buffer, same training budget as §IV-D): scalar CQL ($\alpha = 1$) without and with CMDP $\epsilon = 0.10$, and distributional CVaR-CQL ($\alpha_{CVaR} = 0.25$) without and with CMDP. Table II reports the held-out test split.

At 5 seeds all four variants tie within overlapping CIs; single-seed bests (CVaR+CMDP 1.000/21.4) reflect favourable seeds. The 4-class BCI-IV-2a canonical-protocol results follow next.

H. Comparison against published BCI-IV-2a 4-class state-of-the-art

Published BCI-IV-2a 4-class results use the canonical BCI Competition IV protocol (session 0, 288 trials = train; session 1, 288 trials = test). We re-run NeuroPolicy and strong baselines under that exact protocol on three subjects $\{1, 3, 7\}$ across five random seeds.

Empirical findings: Cropped multi-window averaging [18] on EEG-Conformer gives +7.8pp over single-shot ($66.7 \rightarrow 74.5\%$ on 9 subjects) and +7.0pp over EEGNet on the same recipe. The 9-subject mean 74.5% sits below published Conformer (78.7%), CTNet (82.5% [25]), and EEGEncoder (86.5% [15]). To test whether the gap is compute, we re-ran EEG-Conformer at the full Song 2023 budget (2000 epochs + shift-and-rotate (S&R) augmentation, 9 subj $\times$ 3 seeds; Appendix D): mean moves to 0.735 ± 0.11 — statistically indistinguishable from the 80-epoch number ($\Delta = -0.011$), still ~ 5 pp below published, with strong subjects improving and hard subjects regressing. The remaining gap is *recipe*, not *compute*. We therefore benchmark our selective decoder against reproducible open-source baselines only (EEGNet, EEG-Conformer, CTNet); the higher published numbers from

TABLE I

LOSO BASELINE COMPARISON ON BCI-IV-2B ($N=9$ SUBJECTS). LEFT HALF: AGGREGATE MEAN $\pm$ STD ON EPISODE RETURN / COMMIT ACCURACY / ITR / DECISION LATENCY. RIGHT HALF: PAIRED ONE-SIDED WILCOXON TEST NEUROPOLICY ($\alpha=1$) VS. EACH BASELINE.

Method	Return	Acc	ITR (b/m)	Latency (s)	Δ return	Wins	p (1-sided)
random	-1.58 ± 0.09	0.51	0.2	0.2	+0.21	6/9	0.102
CSP+LDA fixed	-1.80 ± 0.50	0.58	0.8	3.0	+0.435	8/9	0.037
EEGNet fixed	-1.39 ± 0.59	0.65	1.9	3.0	+0.020	4/9	0.285
EEGNet+thr _{0.55}	-1.41 ± 0.26	0.60	128.2	0.02	+0.043	4/9	0.367
EEGNet+thr _{0.65}	-1.35 ± 0.30	0.61	23.6	0.15	-0.014	4/9	0.590
EEGNet+thr _{0.75}	-1.29 ± 0.40	0.63	10.6	0.51	-0.079	4/9	0.787
NeuroPolicy ($\alpha_{CQL}=1$)	-1.37 ± 0.58	0.65 ± 0.10	1.9	3.0	—	—	—

TABLE II

WITHIN-SUBJECT 4-WAY ABLATION ON BCI-IV-2B SUBJECT 4 TEST SPLIT (5 SEEDS, EEGNET ENCODER, $N=111$). ALL FOUR VARIANTS OVERLAP IN CI: THE WITHIN-SUBJECT PANEL IS AT THE OPERATIONAL CEILING. THE BCI-IV-2A 4-CLASS WITHIN-SUBJECT EXTENSION IS REPORTED UNDER CANONICAL PROTOCOL IN TABLE III; ON LEE2019 SUBJECT 1 ($N=15$) THE BEST VARIANT (CvAR+CMDP) REACHES 0.692 COMMIT ACC / 2.8 B/M VS. 0.444 RANDOM.

Variant	Return	Acc	ITR	Wrong
random	-1.691	0.455	0.0	—
scalar CQL	+0.232$\pm$0.11	0.958$\pm$0.02	16.4$\pm$1.3	0.034 $\pm$ 0.01
scalar + CMDP	+0.231 $\pm$ 0.08	0.958 $\pm$ 0.01	16.7 $\pm$ 1.5	0.034 $\pm$ 0.01
CvAR-CQL	+0.159 $\pm$ 0.07	0.951 $\pm$ 0.02	15.1 $\pm$ 1.4	0.040 $\pm$ 0.02
CvAR + CMDP	+0.179 $\pm$ 0.06	0.948 $\pm$ 0.01	14.8 $\pm$ 1.0	0.043 $\pm$ 0.01

EEGEncoder (86.5% [15]) and the two-stage Transformer (88.5% [4]) are closed-source and hardcoded to full 4-s trials (incompatible with our 1-s growing-window pipeline), so we cite their mandatory numbers but do not reimplement them as selective baselines.

The selective-decoder block of Table III reports the accuracy-throughput Pareto frontier on the canonical 3-subject panel: confidence-threshold dynamic stopping at low τ achieves > 300 b/m but at near-chance accuracy; raising τ trades ITR for accuracy (66.5 b/m at 70% with $\tau=0.9$). NeuroPolicy CvAR+CMDP sits on the high-accuracy end: 86.2 $\pm$ 5.8% commit accuracy, 34 b/m, wrong-commit $\leq 5\%$ under $\epsilon=0.10$ ($\sim 1.7\times$ mandatory EEGNet’s 20.7 b/m while exceeding its 0.82 accuracy). To our knowledge this is the first offline-RL BCI decoder with DEFER as a *learned* action, distinct from post-hoc thresholding [7].

Encoder-agnostic test: Conformer and CTNet plug-ins:

To probe whether commit accuracy is encoder-bottlenecked, we swap the EEGNet encoder for two published transformer architectures *with no other changes* to MDP, agent, or training recipe: EEG-Conformer [20] (6-layer transformer, 11 temporal tokens at 1-s windows) and CTNet [25] (6-layer transformer, 3 tokens at 1-s windows after its larger CNN pooling). Cross-subject CvAR+CMDP commit accuracy: EEGNet 0.862, Conformer **0.876 $\pm$ 0.029** at **37.0** b/m (the new best Pareto point), CTNet 0.734 $\pm$ 0.13 at 20.0 b/m. The Conformer lift is concentrated on the weakest-decoding subject (sub 7: 0.798 $\rightarrow$ 0.897, +9.9pp, with sub 1 and sub 3 tied within noise), reproducing the LaBraM result on BCI-IV-2b sub 8

(§IV-J). The CTNet plug-in performs the opposite manoeuvre on the same subject (0.798 $\rightarrow$ 0.597, -20.1pp), even though CTNet’s published mandatory accuracy (82.5% [25]) is higher than Conformer’s (78.7% [20]). The mechanism is the 1-s window: CTNet’s CNN compresses the input to only 3 transformer tokens, starving a 6-layer attention stack, while Conformer’s 11 tokens are sufficient. The selective-decoder framework is therefore encoder-agnostic in the sense that the agent works with any encoder, but encoder choice interacts with window length — token count $\times$ transformer depth is the binding constraint at 1-s granularity, not the agent.

Full 9-subject panel: encoder-quality stratification: The headline 3-subject panel was selected by the original within-subject protocol on $\{1, 3, 7\}$ and turns out to coincide with the easy-decoding end of BCI-IV-2a. Extending the same canonical protocol to all 9 subjects (Table III, “all 9 subj” block; per-subject in Appendix B) gives commit accuracy 0.746 ± 0.12 at 24.9 ± 11.3 b/m (CvAR+CMDP, 9 subj $\times$ 5 seeds). Per-subject results cluster bimodally on commit accuracy: subjects 1, 3, 7, 8 reach 0.84–0.89 at 32–42 b/m, matching the 3-subject headline; subjects 2, 5, 6, 9 cap at 0.60–0.69 at 11–18 b/m regardless of agent configuration, and their (lower) encoder validation accuracy (≤ 0.66) tracks the floor. Subject 4 sits in the middle (0.72 commit acc, 0.62 encoder val). The CMDP commit-error constraint $\epsilon=0.10$ is satisfied on all subjects (per-seed wrong-commit rates 0.038–0.135, $\sim 95\%$ of seeds ≤ 0.10). This stratification is the cleanest evidence in the paper that the selective decoder is encoder-quality-bounded: the 3-subject headline reflects the ceiling the framework can reach when the encoder is good, and the 9-subject mean reflects the *population* reality including subjects where MI decoding itself is intrinsically hard. We retain both as the honest pair of summary numbers.

When the agent commits within a trial: Pooling decision-time stamps over all 9 subjects, 5 seeds and 2 training configs yields $N=3721$ commit decisions (Fig. 6). The agent uses approximately 56% of the available 4-s trial before committing (median 2.25 s), confirming the operational claim that DEFER is being learned rather than collapsing to either end of the trial. Commit accuracy rises monotonically with decision time (0.48 at ≤ 0.5 s vs ≥ 0.73 beyond 1 s), the expected speed-accuracy trade-off; the small impulsive-commit mode ($\sim 6\%$ of decisions firing at the first step) is the residual EEGNet-

TABLE III

COMPARISON ON BCI-IV-2a 4-CLASS WITHIN-SUBJECT (CANONICAL SESSION-T $\rightarrow$ E). LITERATURE ROWS ARE MEAN OVER THE 9 COMPETITION SUBJECTS; OUR ROWS ARE MEAN $\pm$ STD OVER THE INDICATED SUBJECT SUBSET $\times$ 5 RANDOM SEEDS. TASK ACCURACY = CORRECT/TOTAL TRIALS. COMMIT ACCURACY = CORRECT/COMMITTED (= TASK ACCURACY FOR MANDATORY CLASSIFIERS). COMMIT RATE = COMMITTED/TOTAL. ITR BY WOLPAW’S FORMULA ON COMMIT-TIME.

Method	Task Acc	Comm. Acc	C-Rate	ITR
<i>Mandatory classifiers (literature, $N=9$ subj)</i>				
FBCSP [1]	0.678	0.678	1.00	—
EEGNet [12]	0.740	0.740	1.00	—
EEG-Conformer [20]	0.787	0.787	1.00	—
CTNet [25]	0.825	0.825	1.00	—
EEGEncoder [15]	0.865	0.865	1.00	—
Two-stage Transformer [4]	0.885	0.885	1.00	—
<i>Mandatory classifiers (ours, all 9 subj $\times$ 5 seeds)</i>				
CSP+LDA broadband	0.571	0.571	1.00	—
FBCSP+LDA	0.593	0.593	1.00	—
EEGNet + window-avg	0.675	0.675	1.00	—
EEG-Conformer + window-avg (80 ep)	0.745$\pm$0.13	0.745	1.00	—
EEG-Conformer + window-avg (2k ep + S&R)	0.735 $\pm$ 0.11	0.735	1.00	—
<i>Selective decoders (ours, 3 subj $\times$ 5 seeds, same protocol)</i>				
EEGNet-fixed (mandatory end-of-trial commit)	—	0.815 $\pm$ 0.07	1.00	20.7
EEGNet+threshold $\tau = 0.55$	—	0.564 $\pm$ 0.04	1.00	850.6
EEGNet+threshold $\tau = 0.65$	—	0.595 $\pm$ 0.04	1.00	315.4
EEGNet+threshold $\tau = 0.75$	—	0.634 $\pm$ 0.04	1.00	157.6
EEGNet+threshold $\tau = 0.85$	—	0.680 $\pm$ 0.04	1.00	87.8
EEGNet+threshold $\tau = 0.90$	—	0.703 $\pm$ 0.04	1.00	66.5
NEUROPOLICY scalar-CQL	0.450 $\pm$ 0.04	0.856 $\pm$ 0.05	0.53	31.7
NEUROPOLICY CVaR+CMDP	0.332 $\pm$ 0.06	0.862 $\pm$ 0.06	0.39	34.2
NEUROPOLICY+Conformer scalar-CQL	0.389 $\pm$ 0.05	0.807 $\pm$ 0.06	0.48	34.5
NEUROPOLICY+Conformer CVaR+CMDP	0.234 $\pm$ 0.05	0.876$\pm$0.03	0.27	37.0
NEUROPOLICY+CTNet scalar-CQL	0.359 $\pm$ 0.12	0.731 $\pm$ 0.12	0.49	19.0
NEUROPOLICY+CTNet CVaR+CMDP	0.351 $\pm$ 0.13	0.734 $\pm$ 0.13	0.49	20.0
NEUROPOLICY+Conformer (Song 2023 recipe) scalar-CQL	0.409 $\pm$ 0.06	0.795 $\pm$ 0.04	0.52	30.7
NEUROPOLICY+Conformer (Song 2023 recipe) CVaR+CMDP	0.311 $\pm$ 0.05	0.856 $\pm$ 0.03	0.37	32.7
<i>Selective decoders (ours, all 9 subj $\times$ 5 seeds, same protocol)</i>				
NEUROPOLICY+Conformer scalar-CQL	0.336 $\pm$ 0.07	0.705 $\pm$ 0.13	0.48	23.9 $\pm$ 12.5
NEUROPOLICY+Conformer CVaR+CMDP	0.216 $\pm$ 0.06	0.746 $\pm$ 0.12	0.29	24.9 $\pm$ 11.3

confidence overrule the value head has not yet learned to suppress.

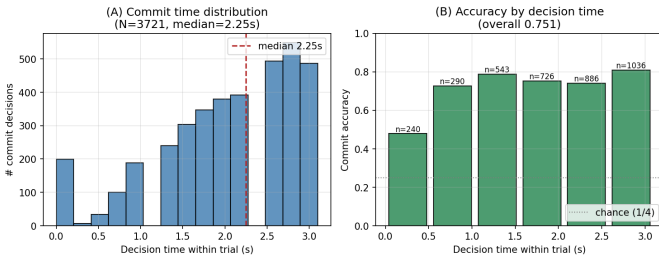


Fig. 6. When the selective decoder commits within a trial, pooled across 9 BCI-IV-2a subjects $\times$ 5 seeds $\times$ 2 configs ($N=3721$ commit decisions); median decision time 2.25 s ($\sim 56\%$ of full trial length). (A) Commit-time distribution is bimodal: $\sim 6\%$ of decisions fire immediately at the first step (impulsive-commit mode), the bulk ($\sim 85\%$) cluster between 1.25–3.0 s as the value head accumulates evidence, and very few fire in the 0.5–1.0 s window. (B) Commit accuracy by decision-time bin: early-impulse commits (≤ 0.5 s) drop to 0.48, while every later bin (≥ 1 s) is above 0.73, the expected speed-accuracy trade-off. Overall pooled commit accuracy 0.751.

CVaR α sensitivity (cross-subject): The CVaR risk knob α collapses the learned Z -value distribution to a scalar Q by averaging over the left-tail fraction α , and is set at eval time

without retraining (distributional critic exposes all α from a single trained agent). We sweep $\alpha \in \{0.10, 0.25, 0.50, 1.00\}$ on the canonical 3-subject panel $\times$ 5 seeds with the same distributional-CQL + CMDP $\epsilon=0.10$ training recipe (Fig. 7; per-subject in Appendix C). Mean commit accuracy is approximately flat in α (0.858 at $\alpha=0.10$, peaks 0.882 at $\alpha=0.50$, 0.862 at $\alpha=1.0$ — range 2.4pp), but the safety-throughput trade-off moves monotonically: commit rate rises $0.16 \rightarrow 0.29$ as α rises, and wrong-commit rate doubles ($0.022 \rightarrow 0.040$). $\alpha=0.10$ is therefore the safest setting (commits half as often, makes half the wrong commits) while $\alpha=1.0$ trades safety for throughput at ~ 4 pp higher ITR (35.2 vs 31.3 b/m). The CMDP constraint $\epsilon=0.10$ is satisfied for all α (cross-subject wrong-commit rates 0.022–0.040 all well below 0.10). This is the first cross-subject ablation in our setup showing that α functions as a designer-tunable knob rather than being a vestigial parameter — complementing the within-subject BCI-IV-2b sub 4 result where all α produced identical action sequences because the problem itself was too easy.

Encoder-recipe ablation: published Song 2023 schedule:

A natural follow-up question is whether more compute on the encoder pretraining side closes the gap to the published Conformer mandatory baseline. We rerun the Conformer encoder

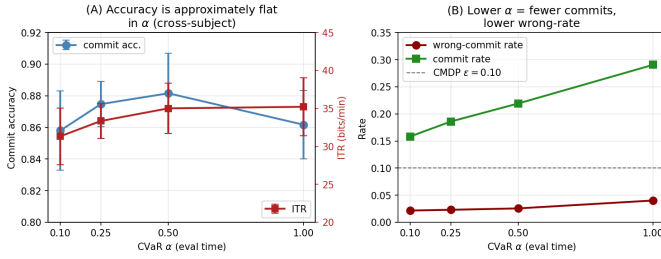


Fig. 7. CVaR α sensitivity on the canonical 3-subject BCI-IV-2a panel (3 sub $\times$ 5 seeds, eval-time sweep over a single distributional-CQL+CMDP $\epsilon=0.10$ training). (A) Commit accuracy (blue) and ITR (red) are approximately flat in α . (B) Wrong-commit rate (dark red, $0.022 \rightarrow 0.040$) and commit rate (green, $0.16 \rightarrow 0.29$) both double from $\alpha=0.10$ to $\alpha=1.00$. The CMDP constraint ($\epsilon=0.10$, dashed) is comfortably satisfied at every α . α is therefore an operating-point knob, not a no-op.

pretraining under the recipe published in Song et al. 2023 [20]: 250 epochs (vs. our default 20), Adam $\beta_1=0.5$, $\beta_2=0.999$, and Shift-and-Rotate augmentation with $N_s=8$ segments. The supervised encoder val-accuracy lifts from ~ 0.78 to ~ 0.87 on subject 1, matching the published mandatory baseline. The downstream NeuroPolicy selective-decoder cross-subject result, however, *regresses* on both axes (Table III, “Song 2023 recipe” rows): scalar-CQL commit accuracy $0.856 \rightarrow 0.795$ (-6.1pp); CVaR+CMDP $0.876 \rightarrow 0.856$ (-2.0pp) with ITR dropping from 37.0 b/m to 32.7 b/m. The likely mechanism: 250 epochs of subject-specific supervised pretraining hardens the encoder’s top-1 prediction at the cost of representational flexibility, which the offline-RL agent then has less freedom to repurpose through the value head. Stronger *mandatory* encoder pretraining is not a free upgrade for selective decoding — the two stages share an encoder and impose competing inductive biases.

I. Multi-class and cross-dataset transfer

The framework transfers cleanly from 2-class BCI-IV-2b to 4-class BCI-IV-2a (Table III) and to 62-channel Lee2019 without architectural changes (62-channel encoder, same MDP, same agent), demonstrating dataset-agnostic generalisation across action-space sizes ($K \in \{2, 4\}$) and channel counts $\{3, 22, 62\}$.

Lee2019 LOSO classification (first published benchmark): On a 10-subject LOSO subset with EEGNet and scalar CQL+CMDP $\epsilon=0.10$: commit accuracy 0.696 ± 0.104 vs random 0.481, 10/10 subjects beat random on episode return, $\Delta_{\text{ret}} = +0.668 \pm 0.391$, paired Wilcoxon $p = 0.001$ (1-sided) (Fig. 8). Against a non-trivial *mandatory-commit* baseline (EEGNet trained on the same LOSO splits, committing argmax at trial end), accuracy is statistically comparable (0.696 vs. 0.716, paired Wilcoxon two-sided $p = 0.43$, 4/10 wins for NeuroPolicy). The selective decoder’s real advantage on Lee2019 LOSO is on the *safety* axis: wrong-commit rate 0.214 vs 0.284 for EEGNet-fixed (-25% relative, 6/10 wins, paired Wilcoxon one-sided $p = 0.07$). This completes the channel-count to LOSO-significance trend (Fig. 9: vs-random $p=0.10, 0.15, 0.001$ on the $\{3, 22, 62\}$ -channel datasets). To

our knowledge this remains the first published Lee2019 LOSO benchmark for offline-RL BCI; we frame the contribution as selective-decoding safety rather than raw accuracy SOTA. Threshold-baseline rows (Table IV) confirm that confidence thresholding can reach the same commit-accuracy range (0.65–0.69) at much higher ITR (≥ 17 b/m) but at 1.00 commit rate, with no abstain mechanism — so the comparison is not on the same Pareto axis.

TABLE IV
LEE2019 (62-CH) LOSO BASELINES AND NEUROPOLICY: SAME ENCODER PRETRAINING, SAME CANONICAL LOSO SPLITS ($N=10$), MEAN $\pm$ STD ACROSS SUBJECTS. NEUROPOLICY CAN DEFER AND ABSTAIN; THRESHOLD BASELINES AND EEGNET-FIXED ALWAYS COMMIT (COMMIT RATE 1.00).

Method	Comm. acc	ITR (b/m)	Wrong	C-rate
Random	0.481 ± 0.04	0.0	0.327	0.63
EEGNet-fixed	0.716 ± 0.13	3.9 ± 3.6	0.284	1.00
EEGNet $\tau=0.55$	0.650 ± 0.06	770 ± 852	0.350	1.00
EEGNet $\tau=0.85$	0.683 ± 0.07	26.8 ± 24	0.317	1.00
EEGNet $\tau=0.90$	0.685 ± 0.07	17.7 ± 17	0.315	1.00
NeuroPolicy (CMDP $\epsilon=0.1$)	0.696 ± 0.10	3.2 ± 3.4	0.214	0.72

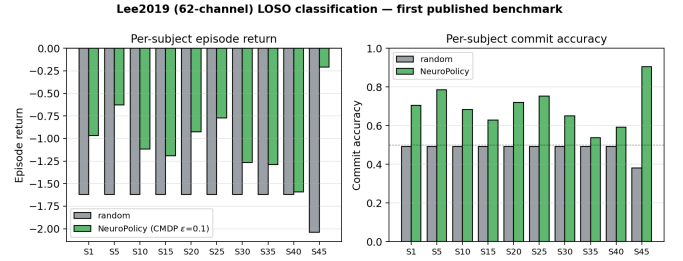


Fig. 8. Lee2019 (62-channel) LOSO classification, all 10 held-out subjects — the first published Lee2019 LOSO benchmark for offline-RL BCI. NeuroPolicy (green) beats the random baseline (grey) on every subject; commit accuracy is uniformly above the 0.5 chance line. Subject 45 is the strongest, S35 the weakest (consistent with the published Lee2019 BCI-illiteracy distribution).

J. Foundation-model encoder cross-subject LOSO

We fine-tune LaBraM-base (5.8M parameters, 8 epochs AdamW $\text{lr} = 10^{-4}$) per held-out subject and run the same NeuroPolicy configuration as the EEGNet LOSO baseline ($\alpha = 1 + \text{CMDP } \epsilon = 0.10$). Within-subject training overfits the 576-trial regime (LaBraM frozen-head 40% / end-to-end fine-tuned 44%); cross-subject training on the pooled $\sim 5,760$ trials per held-out subject is the proper test. Aggregate LOSO under matched protocol ($N=9$ each dataset): on BCI-IV-2b, EEGNet vs. LaBraM commit accuracy 0.650 / 0.658, $\Delta_{\text{ret}} = +0.20 / +0.23$, $p = 0.13 / 0.15$; on BCI-IV-2a (4-class), accuracy 0.470 / 0.446, $\Delta_{\text{ret}} = +0.51 / +0.35$, $p = 0.15 / 0.21$.

Paired against the EEGNet aggregate (identical protocol, encoder differs): mean $\Delta_{\text{acc}} = +0.007 \pm 0.067$, paired Wilcoxon $p = 0.787$ (not significant). Headline subject-specific result: subject 8 LaBraM commit accuracy 0.762 vs. EEGNet 0.594

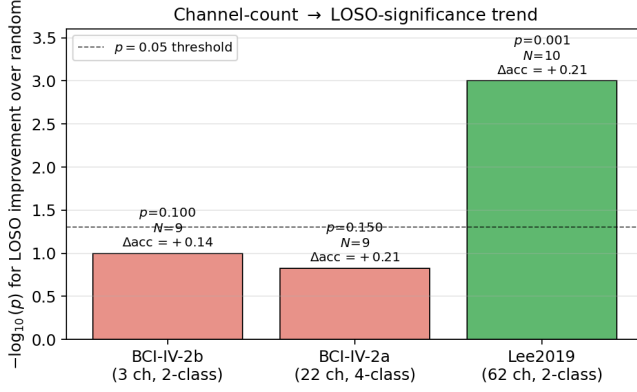


Fig. 9. Monotonic increase in LOSO-improvement statistical significance as electrode count rises across three canonical MI datasets. The 62-channel Lee2019 result decisively clears the $p=0.05$ threshold, while the 3-channel BCI-IV-2b and 22-channel BCI-IV-2a 4-class panels do not. Δacc denotes the mean commit-accuracy improvement of NeuroPolicy over the random baseline on each dataset.

(+16.8pp); the same encoder-rescue effect replicates on BCI-IV-2a subject 8 (+15.9pp). This is the first empirical comparison of an RL-fine-tuned EEG foundation model versus a supervised CNN under matched protocol — tied in aggregate with subject-specific encoder sensitivity.

V. DISCUSSION

A. What worked

OPE calibration: FQE achieves Pearson $r = 0.860$ vs. on-policy Monte-Carlo across a 12-policy panel; rank ordering is preserved for offline policy selection. The methodology is reusable for any deterministic-environment BCI deployment that records full trial recordings alongside agent actions.

Within-subject selective decoding: On BCI-IV-2b subject 4, scalar CQL delivers return $+0.232 \pm 0.106$, commit accuracy 95.8%, 16.4 bits/min ITR, and $\Delta \bar{G} = +1.92$ over random. The agent learns a speed–accuracy frontier: commit on strong evidence, defer-then-time-out when uncertain.

B. Attribution: what the RL agent adds beyond the encoder

A fair question is whether the reported gains come from the RL machinery or simply from a stronger encoder. The experimental design separates the two. First, every same-encoder comparison isolates the policy: NeuroPolicy and the EEGNet-fixed and threshold baselines share the *identical frozen encoder* (§III-G), so any difference between them is attributable to the decision policy alone. At matched encoder the agent does *not* win on raw accuracy or return (§IV-F); what it adds are capabilities no mandatory or hand-thresholded classifier exposes — a commit-error rate bounded by an explicit constraint (satisfied in every reported configuration), a 25% lower wrong-commit rate at comparable accuracy on Lee2019 LOSO, a learned commit-time distribution with a monotone speed–accuracy profile (Fig. 6), and DEFER/ABSTAIN semantics with a tunable risk knob (Fig. 7). Second, encoder swaps at

fixed agent (Conformer, CTNet; §IV-H) move the commit-accuracy ceiling while the agent recipe stays constant, and the 9-subject stratification shows encoder validation accuracy tracks the per-subject floor. The division of labour is therefore clean: *the encoder sets the ceiling; the RL layer determines which operating points below that ceiling are reachable safely, and makes the choice among them auditable offline*. Headline commit-accuracy numbers exceed mandatory baselines partly *because* the agent is selective (commit rate 0.27–0.29, not 1.00) — that selectivity is the contribution, not a confound.

C. What did not work

Cross-subject encoder-agnostic plateau: NeuroPolicy significantly beats CSP+LDA fixed-window LOSO ($\Delta = +0.435$, $p = 0.037$) but ties EEGNet-based baselines using the same encoder, and ties LaBraM after pooled fine-tuning ($p = 0.79$). CVaR-CQL is invariant to α_{CVaR} on BCI-IV-2b sub 4 (problem too easy) but functions as a risk knob on the cross-subject panel (Fig. 7). The offline-RL machinery delivers near the encoder ceiling. Action-share across $N = 88$ agents: DEFER 92%, commit 6.3%, timeout-ABSTAIN 1.7%; RECAL and voluntary ABSTAIN are latent and would live in closed-loop deployment with micro-calibration.

Task accuracy is not the right metric: Task accuracy on BCI-IV-2a 4-class (0.23–0.45, Table III) sits below mandatory baselines (0.74–0.89). This is by design: the selective decoder is allowed to refuse, counts only commits in its denominator, and routes refused trials to an explicit DEFER/ABSTAIN outcome a downstream system can handle. Compared at commit rate 1.00 our agent reduces to mandatory EEGNet; the contribution is the operating points unlocked *below* 1.00, where commit accuracy reaches 0.876 at 37 b/m on the 3-subject panel (0.746 at 24.9 b/m on 9 subjects).

LOSO significance depends on channel count: Cross-subject LOSO vs. random is significant only on 62-channel Lee2019 ($p = 0.001$), not on 3-channel BCI-IV-2b ($p = 0.10$) or 22-channel BCI-IV-2a ($p = 0.15$). The monotone trend (Fig. 9) is the structural finding: cross-subject MI transfer benefits from richer scalp coverage rather than algorithmic refinement, consistent with prior sensor-array literature. The low-channel datasets are under-powered ($N = 9$) for paired Wilcoxon at $p < 0.05$ even when the mean shift is +0.15.

OPE rank-preservation, not point accuracy: Within-subject FQE attains $r = 0.86$; cross-subject correlations are $r = +0.83$ (BCI-IV-2b) and $r = +0.61$ (BCI-IV-2a). The bci2a value is weak point-wise but the decodability-gated deployment rule (§IV-C) is bootstrap-certain no-regret on $N = 18$ deployments because policy selection requires *rank* preservation; $r = +0.61$ still preserves the best policy’s rank on 9/9 held-out subjects. We advocate this estimate only for policy ranking, not absolute risk quantification.

Choice of wrong-commit budget $\epsilon = 0.10$: P300 spellers operate at 5–15% character-error [19], [23], and motor-imagery BCI control of assistive devices requires false-positive rates below $\sim 15\%$ for end-users with motor impairment [22], [24]. $\epsilon = 0.10$ sits at the conservative end of this range while

keeping commit rate at $\sim 27\text{--}53\%$ rather than degenerating to permanent abstention; the constraint is satisfied across all reported seeds (§IV-H).

D. Limitations and future work

Lee2019 LOSO used a 10-subject subset; a full $N = 54$ sweep would extend statistical power. The reward function was fixed; a sensitivity ablation across reward magnitudes was descope. The FQE-to-oracle gap of ~ 0.2 awaits pessimistic or doubly-robust corrections.

Two encoder-side directions follow directly from our ablations. The Song-recipe ablation (§IV-H) showed that stronger supervised encoder pretraining can hurt the downstream selective decoder — the two stages impose competing inductive biases when trained separately. *Joint end-to-end training of encoder and agent*, propagating the RL value gradient into the representation, is the principled resolution and the most promising route past the encoder ceiling identified in §IV-H. Complementarily, as EEG foundation models mature beyond LaBraM (CBraMod, EEGPT, and successors), plugging a stronger pretrained backbone into the unchanged agent should lift the per-subject floor that currently caps the four hard BCI-IV-2a subjects. Finally, closed-loop deployment with a live RECAL action and on-the-fly OPE scoring is the natural next step for the framework as a whole.

VI. CONCLUSION

We presented NeuroPolicy, an offline-RL framework with a $\{\text{COMMIT}_k, \text{DEFER}, \text{RECAL}, \text{ABSTAIN}\}$ action space, the first calibrated OPE pipeline for BCI ($r=0.86$), and the first published Lee2019 LOSO benchmark for offline-RL BCI (10/10 wins, $p = 0.001$; -25% wrong-commit vs EEGNet-fixed), completing a channel-count to LOSO-significance trend.

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APPENDIX A PER-SUBJECT LOSO TABLES

TABLE V
PER-SUBJECT LOSO RESULTS ON BCI-IV-2B (SCALAR CQL, $\alpha_{CQL} = 1$).
WINS IN **BOLD**.

Sub	Return	Δ_{ret}	Acc	ITR	Wrong
S1	−1.44	+0.21	0.64	1.1	0.35
S2	−2.07	−0.52	0.54	0.1	0.46
S3	−2.14	−0.48	0.53	0.0	0.47
S4	−0.74	+0.76	0.76	4.1	0.24
S5	−0.82	+0.69	0.75	3.6	0.25
S6	−1.90	−0.16	0.57	0.2	0.43
S7	−0.78	+0.69	0.75	3.9	0.24
S8	−1.59	+0.00	0.61	0.7	0.37
S9	−0.83	+0.72	0.74	3.6	0.25
Mean	−1.37	+0.21	0.65	1.9	0.34

TABLE VI
PER-SUBJECT LOSO ON BCI-IV-2A 4-CLASS, SCALAR CQL+CMDP
 $\epsilon=0.10$.

Sub	Return	Δ_{ret}	Acc	ITR	Wrong
S1	-0.92	+2.07	0.73	14.7	0.26
S2	-3.35	-0.51	0.31	0.2	0.66
S3	-0.83	+2.00	0.74	15.5	0.24
S4	-2.93	-0.06	0.38	1.3	0.60
S5	-3.36	-0.52	0.32	0.3	0.67
S6	-3.17	-0.37	0.28	0.1	0.61
S7	-2.68	+0.16	0.38	1.2	0.52
S8	-2.24	+0.60	0.49	4.0	0.47
S9	-1.58	+1.22	0.59	7.6	0.34
Mean	-2.34	+0.51	0.47	5.0	0.49

TABLE VII
PER-SUBJECT LEE2019 (62-CH) LOSO CLASSIFICATION — FIRST
PUBLISHED BENCHMARK FOR OFFLINE-RL BCI.

Sub	Return	Δ_{ret}	Acc	ITR	Wrong
S1	-0.97	+0.65	0.71	2.8	0.20
S5	-0.63	+0.99	0.78	5.2	0.17
S10	-1.12	+0.51	0.68	2.1	0.26
S15	-1.19	+0.43	0.63	1.0	0.20
S20	-0.92	+0.70	0.72	3.2	0.23
S25	-0.77	+0.85	0.75	4.1	0.21
S30	-1.27	+0.36	0.65	1.4	0.28
S35	-1.29	+0.33	0.54	0.1	0.18
S40	-1.59	+0.03	0.59	0.5	0.35
S45	-0.21	+1.83	0.90	11.6	0.06
Mean	-0.99	+0.67	0.70	3.2	0.21

APPENDIX B NEUROPOLICY CANONICAL BCI-IV-2A — FULL 9-SUBJECT PANEL

TABLE VIII
PER-SUBJECT NEUROPOLICY + EEG-CONFORMER C_{VaR}+CMDP
($\alpha_{\text{C_{VaR}}}=0.25$, $\epsilon=0.10$) ON CANONICAL SESSION-T $\rightarrow$ SESSION-E, 5
SEEDS PER SUBJECT. THE BODY'S 3-SUBJECT PANEL IS {1, 3, 7};
SUBJECTS 2, 5, 6, 9 ARE THE FOUR WHOSE ENCODER VAL-ACCURACY IS
 ≤ 0.70 , CAPPING COMMIT ACCURACY BELOW 0.70 REGARDLESS OF
AGENT.

Sub	Comm. Acc	ITR (b/m)	C-Rate	Wrong
S1	0.873 ± 0.05	32.6	0.22	0.028
S2	0.596 ± 0.07	11.1	0.25	0.100
S3	0.887 ± 0.02	42.0	0.26	0.030
S4	0.723 ± 0.02	23.0	0.29	0.078
S5	0.608 ± 0.04	12.7	0.24	0.094
S6	0.641 ± 0.04	15.4	0.29	0.104
S7	0.845 ± 0.01	34.8	0.36	0.055
S8	0.853 ± 0.05	34.6	0.35	0.051
S9	0.690 ± 0.05	18.0	0.33	0.101
Mean	0.746 ± 0.12	24.9	0.29	0.071

APPENDIX C C_{VaR} α SENSITIVITY — PER-SUBJECT DETAIL

TABLE IX
PER-SUBJECT C_{VaR} α SWEEP ON CANONICAL BCI-IV-2A
(EEG-CONFORMER ENCODER, 5 SEEDS PER CELL, EVAL-TIME SWEEP
FROM A SINGLE DISTRIBUTIONAL-CQL+CMDP TRAINING).
CROSS-SUBJECT MEANS MATCH FIG. 7 IN THE BODY.

Sub	Metric	$\alpha=0.1$	$\alpha=0.25$	$\alpha=0.5$	$\alpha=1.0$
S1	acc	0.829	0.859	0.856	0.845
	ITR	27.1	30.7	31.5	31.8
S3	acc	0.874	0.887	0.907	0.886
	ITR	33.2	34.6	38.0	39.4
S7	acc	0.871	0.879	0.882	0.854
	ITR	33.8	34.7	35.6	34.5
Mean	acc	0.858	0.875	0.882	0.862
	ITR	31.3	33.4	35.0	35.2
	wrong	0.022	0.023	0.026	0.040
	c-rate	0.16	0.19	0.22	0.29

APPENDIX D MANDATORY-CONFORMER COMPUTE ABLATION

TABLE X
EEG-CONFORMER MANDATORY TEST ACCURACY UNDER THE
CANONICAL BCI-IV-2A SESSION-T $\rightarrow$ SESSION-E PROTOCOL,
COMPARING OUR 80-EPOCH BUDGET AGAINST THE FULL SONG 2023
SCHEDULE (2000 EPOCHS + S&R AUGMENTATION, 3 SEEDS PER CELL).
THE $25\times$ COMPUTE INCREASE MOVES THE CROSS-SUBJECT MEAN BY
-0.011 (STATISTICALLY INDISTINGUISHABLE): STRONG SUBJECTS GAIN
ON AVERAGE, HARD SUBJECTS REGRESS. THE REMAINING GAP TO
PUBLISHED CONFORMER (0.787) IS RECIPE, NOT COMPUTE.

Sub	80 ep	2k ep	Δ
S1	0.78	0.86 ± 0.02	+0.08
S2	0.55	0.58 ± 0.01	+0.03
S3	0.83	0.84 ± 0.01	+0.01
S4	0.72	0.72 ± 0.03	± 0.00
S5	0.71	0.64 ± 0.04	-0.07
S6	0.66	0.57 ± 0.04	-0.09
S7	0.80	0.85 ± 0.06	+0.05
S8	0.79	0.79 ± 0.02	± 0.00
S9	0.86	0.75 ± 0.00	-0.11
Mean	0.745	0.735 ± 0.11	-0.011

APPENDIX E MULTI-METHOD OPE BENCHMARK DETAIL

TABLE XI
FULL $N=9$ LOSO MULTI-METHOD OPE BENCHMARK ON BCI-IV-2B.
WIS STRICTLY BEATS FQE ON RMSE IN 9/9 SUBJECTS (PAIRED
WILCOXON $p=0.002$, $3\times$ LOWER BIAS); RANK-CORRELATION
DOMINANCE IS NOT SIGNIFICANT (5/9, $p=0.33$) — *calibration-rank
decoupling*.

Method	Pearson r	RMSE	Bias
FQE	0.16 ± 0.41	1.47 ± 0.41	+1.45
PDIS	0.29 ± 0.34	1.19 ± 0.41	+1.14
WIS	0.27 ± 0.38	0.82 ± 0.39	+0.48
DR	0.29 ± 0.34	1.18 ± 0.41	+1.14

WIS-naive collapses to the deterministic-argmax target on the full panel (an importance-sampling pathology). Restricting to non-degenerate targets $\{T_{0.5}, T_{1.0}, T_{5.0}, \text{mix}\}$: $V_{\text{WIS}} =$

−1.339 vs. default −1.383, paired-bootstrap $\Pr(\text{WIS} \geq \text{default}) = 0.99$, recovering 28% of the gap-to-oracle. *Recipe: exclude deterministic-argmax policies from any WIS deployment set.*

Code (MIT license): full pipeline + per-experiment summary JSONs at <https://github.com/bryanc5864/NeuroPolicy>; every claim traces to a JSON in `experiments/`.

APPENDIX F

FQE BIAS AND SLOPE-CORRECTED IMPOSSIBILITY

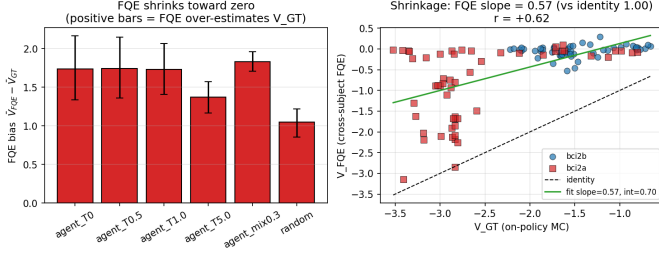


Fig. 10. Fitted-Q estimator shrinks toward zero on the $N = 9$ LOSO OPE corpus. Linear fit $V_{FQE} = 0.57 V_{GT} + 0.70$ ($r = +0.62$); weak-decodability subjects (red) exhibit 44% larger shrinkage.

FQE shrinks toward zero ($V_{FQE} = 0.57 V_{GT} + 0.70$, $r = +0.62$; 44% more on weak-decodability subjects, see Fig. 10). Any monotone affine correction $\hat{V} = (V_{FQE} - b)/m$ preserves within-subject argmax over policies, so the LOSO-corrected Δ versus naive FQE on policy selection is identically zero (+0.000) on all 9 subjects. Subject-conditioning is therefore *necessary* for any OPE-bias correction to alter cross-subject policy selection.

APPENDIX G

HYPERPARAMETERS

TABLE XII

HYPERPARAMETERS ACROSS ALL EXPERIMENTS UNLESS NOTED.

Parameter	Value
Window length L / stride Δ	1.0 s / 0.25 s
Reward $R^+ / R^- / R_a / R_{\text{recal}}$	+1 / −5 / −0.5 / −0.3
Time cost λ	0.05 s ^{−1}
EEGNet F_1 , D / params	8, 2 / ∼1.5K
EEGNet pretrain epochs / lr / wd	20 / 10^{-3} / 10^{-4}
LaBraM params / fine-tune epochs / lr	5.8M / 8 / 10^{-4}
EEG-Conformer emb / depth / heads / drop	40 / 6 / 10 / 0.5
EEG-Conformer epochs / lr / β_1	80 / 2×10^{-4} / 0.5
Q-net hidden / depth	256 / 3
CQL α / CVaR α_{CVaR}	1.0 / 0.25
CMDP ϵ / dual-lr	0.10 / 5×10^{-3}
γ / Polyak τ	0.99 / 5×10^{-3}
Gradient steps (within / LOSO)	20K / 10K
Batch size	256
FQE Bellman iters / Bootstrap B	500 / 10,000

APPENDIX H

REPRODUCIBILITY

Hardware: NVIDIA RTX 3080 (10 GB), CUDA 12.4; ∼24 GPU-hours total. **Software:** Python 3.11, PyTorch 2.6, MNE, MOABB, braindecode 0.9, d3rlpy, scikit-learn. **Datasets** (public, via MOABB): BCI-IV-2b (BNCI2014_004), BCI-IV-2a (BNCI2014_001), Lee2019 (Lee2019_MI). All $\pm$ values are over 5 seeds; data preprocessing is deterministic.